Inorganic Cumulative Examination

February 6, 2009

The NO_3 radical was the topic of a recent seminar and proved to be a deceptively complex molecule. Nitric acid and the anion NO_3- might be viewed as the parents to the radical and they , unlike the radical, are commonly encountered species in chemistry.

1. Provide a practical laboratory scheme for the preparation of nitric acid.
2. What is meant by the term "fuming nitric acid"?
3. How does one prepare the "aqua regia"?
4. Give the Lewis dot representations of HNO_3, NO_3^- and NO_3 radical. Justify the locations of any unpaired electrons and relative the contributions of potential resonance structures.
5. Using VSEPR concepts determine the geometry of the three and the appropriate point group.
6. Much of the seminar revolved around the symmetry and the impact of the symmetry on the vibrational spectrum, i.e., did the molecule have D_{4h} symmetry or C_{2v} symmetry? Describe specifically how vibrational spectroscopy would differentiate between the two symmetries in the case of this radical.
7. Provide an example of a familiar ester formed with nitric acid.
8. Provide a practical laboratory scheme for the preparation of NO.
9. Provide a practical laboratory scheme for the preparation of NO_2.
10. Provide a practical laboratory scheme for the preparation of N_2O.